VIETNAM NATIONAL UNIVERSITY, HANOI
INTERNATIONAL SCHOOL



FINAL REPORT

ECONOMETRICS

PROJECT 7

Lecturer : M.A. Lê Văn Đạo
Ph.D. Nguyen Viet Cuong
Ph.D. Tran Quang Tuyen

Class : INS304901

Group : 7

Members : Lưu Văn Toàn
Mạc Hải Nam
Tạ Quang Khôi
Vũ Ngọc Huyền
Nguyễn Diệu Huyền

Hanoi, June 2026

Name		Tasks	Contribution
Lưu Văn Toàn		Model Specification, Stata Code	20%
Mạc Hải Nam		Tests on Interaction and Quadratic Terms, Graphical presentation and regression output and Stata Code	20%
Tạ Quang Khôi		Model Assumptions and Diagnostic Tests, Regional Wage Differentials	20%
Vũ Ngọc Huyền		Main findings and policy recommendations, Report	20%
Nguyễn Diệu Huyền		Literature Review, Overview dataset and Descriptive statistics	20%

INTERNATIONAL SCHOOL
VIETNAM NATIONAL UNIVERSITY, HANOI

Tables of Contents

1. Introduction 1

2. Literature Review 1

2.1 Human capital theory and wage determination 1

2.2 Education, experience, and earnings 1

2.3 Gender wage gap and marital status 1

2.4 Public–private wage differentials 2

2.5 Regional and urban–rural wage disparities 2

3. Research Design 2

3.1 Overview dataset 2

3.2 Characteristics descriptive statistics 3

3.3 Design model 4

3.3.1 Economic and econometric model of wage determination 4

3.3.2 Assumptions of the multiple linear regression (MLR) model 5

4. Results 8

4.1 Interaction term between gender and public sector (Model 2) 10

4.2 Returns to education: Urban vs Rural workers (Model 3) 10

4.3 Non-linear relationship between experience and log(wage) (Model 4) 11

4.4 Regional Wage Differentials 12

5. Conclusion and Policy 13

5.1 Conclusion 13

5.2 Policy implications 13

6. References 14

INTERNATIONAL SCHOOL
VIETNAM NATIONAL UNIVERSITY, HANOI

List of Tables

Table 1: Variables Explanation 3

Table 2: Descriptive Statistics 4

Table 3: Baseline OLS Regression (Model 1) 6

Table 4: Variance Inflation Factor (VIF) 7

Table 5: Baseline Regression with Robust Standard Errors (Model 1 - Robust) 8

Table 6: Regression Results 9

Table 7: Proportional Wage Differentials by Region 12

INTERNATIONAL SCHOOL
VIETNAM NATIONAL UNIVERSITY, HANOI

List of Figures

Figure 1: Predicted relationship between experience and ln(wage).....11

1. Introduction

Vietnam's rapid economic growth has contributed to rising living standards and poverty reduction. Nevertheless, significant wage disparities persist across regions, sectors, and demographic groups, posing challenges to inclusive growth and labor market development (World Bank, 2022). Understanding these disparities is essential for designing effective labor market policies. This study aims to answer the following research question: What factors determine wage differentials among Vietnamese workers?

Using data from 178,340 individuals in the 2021 Labour Force Survey, this study examines how education, work experience, gender, marital status, employment sector, and location influence hourly wages in Vietnam. The findings provide evidence to support policies aimed at reducing wage inequality and expanding economic opportunities.

2. Literature Review

2.1 Human capital theory and wage determination

Human capital theory posits that wages reflect labor productivity, driven by education and experience (Becker, 1964). (Mincer, 1974) formalizes this through an earnings function linking wages to schooling and experience, forming the foundation of empirical wage regressions.

2.2 Education, experience, and earnings

Education increases adaptability and productivity, yielding positive wage returns globally. Similarly, work experience boosts early-career earnings through skill accumulation, but eventually faces diminishing returns, forming a concave (quadratic) relationship with wages.

2.3 Gender wage gap and marital status

Gender wage gaps persist due to occupational segregation and discrimination (Blau & Kahn, 2017), often remaining significant even when controlling for observables. Concurrently,

marital status influences earnings through household specialization; married individuals frequently receive wage premiums linked to perceived job stability (Becker, 1985).

2.4 Public–private wage differentials

Public sector wages are generally more compressed and less responsive to individual productivity but offer superior job stability (Gregory & Borland, 1999). In transition economies, public sector workers—particularly women—often earn premiums over their private sector counterparts, motivating our interaction analysis.

2.5 Regional and urban–rural wage disparities

Agglomeration effects in urban hubs generate higher productivity and wages compared to rural areas. In Vietnam, the process of economic decentralization has contributed to uneven regional development, as provinces differ in their capacity to attract investment, develop infrastructure, and create employment opportunities. These differences contribute to persistent regional wage disparities, making regional control variables essential for isolating geographic wage premiums from individual human capital effects (Vu, 2011).

3. Research Design

3.1 Overview dataset

The analysis utilizes hourly wage (thousand VND) and its transformed logarithm (*log_wage = ln(wage + 1)*).

Variable	Description
Education (edu)	Number of schooling years
gender	1 = male; 0 = female
married	1 = married; 0 = single
public	1 = working in public sector; 0 = working in private sector
exper	Potential work experience (years)

Variable	Description
urban	1 = living in urban areas; 0 = living in rural areas
region	1=Northern Midlands & Mountains; 2=Red River Delta; 3=Central Coast; 4=Central Highlands; 5=Southeast; 6=Mekong Delta
wage	Hourly wage in thousand VND
ln(wage)	Natural logarithm of hourly wage

Table 1: Variables Explanation

The table below reports the descriptive statistics of the variables used in the analysis. All 178,340 observations are retained. Using ln(wage+1) ensures the transformation is defined even when wage = 0. This avoids undefined logarithmic transformations. The dataset contains both the level of hourly wages (wage, in thousand VND) and its natural logarithm (ln(wage)).

For the baseline OLS model and MLR assumptions, we use wage as required. For proportional interpretations and non-linear patterns, we later work with ln(wage). Although the average hourly wage is 39.14, the large standard deviation and wide range indicate that wages are highly right-skewed. Therefore, taking the logarithm of wages helps reduce the influence of extreme values and allows the regression coefficients to be interpreted approximately as percentage changes.

3.2 Characteristics descriptive statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
log_wage	178,340	3.5316	0.5675	0.0000	8.6715
wage	178,340	39.1361	33.4095	0.1666	5833.33
Education (edu)	178,340	11.4383	4.6899	0	22
exper	178,340	19.8450	12.2086	0	77
gender	178,340	0.5786	0.4938	0	1
public	178,340	0.1894	0.3918	0	1

Variable	Obs	Mean	Std. Dev.	Min	Max
married	178,340	0.7952	0.4036	0	1
urban	178,340	0.4939	0.5000	0	1
region	178,340	3.2284	1.7208	1	6

Table 2: Descriptive Statistics

On average, workers have 11.43 years of education and 19.85 years of work experience, indicating a diverse labor force in terms of human capital. Regarding demographic characteristics, 57.9% of workers are male, 18.9% are employed in the public sector, 79.5% are married, and 49.4% live in urban areas. The regional variable is categorical and takes values from 1 to 6 to represent different geographical regions.

3.3 Design model

3.3.1 Economic and econometric model of wage determination

Wages reflect the marginal productivity of labor and are influenced by human capital, individual characteristics, job characteristics, and labor market conditions. The baseline wage regression is specified as: $wage_i = \beta_0 + \beta_1 edu_i + \beta_2 gender_i + \beta_3 married_i + \beta_4 public_i + \beta_5 exper_i + \beta_6 urban_i + \sum \gamma_j region_j + u_i$ (Region 1 is the omitted base category). In subsequent models, *log_wage* is used to interpret proportional effects and non-linear interactions.

Explanation of explanatory variables and expected signs: - Education (edu): Formal human capital investment that enhances productivity. According to human capital theory, education enhances workers' productivity. Expected sign: (+) - Gender (gender): Controls for gender wage differentials (1 = male). Expected sign: (+) - Marital status (married): Captures job stability and household specialization. Married workers often exhibit greater job stability and stronger work incentives. Expected sign: (+) - Public sector (public): Captures public vs. private wage differences. Expected sign: Ambiguous (+/-) - Work experience (exper):

Reflects on-the-job skill accumulation. Expected sign: (+) - Urban residence (urban): Proxies for labor-market agglomeration effects. Urban areas typically offer higher productivity jobs. Expected sign: (+) - Region (region): Controls for spatial economic disparities relative to the Northern Midlands base region. More developed regions are expected to offer higher wages compared to the base region.

Note on the Age variable: Although *age* is theoretically an important demographic characteristic mentioned in the economic framework, it is intentionally excluded from the econometric model due to perfect multicollinearity. Based on the Mincer equation, potential experience is calculated as $exper = age - edu - 6$ (or $age = exper + edu + 6$). Including age, education, and experience simultaneously would violate the MLR.3 assumption. Thus, the model captures the effect of age entirely through the education and experience variables.

3.3.2 Assumptions of the multiple linear regression (MLR) model

To ensure that the regression coefficients are estimated in an unbiased and reliable manner, the model must satisfy basic assumptions. As a first step, we estimate the baseline wage equation using ordinary least squares:

wage	Coefficient	Std. Err.	t	P>|t|	[95% Conf. Interval]
Education (edu)	2.2241	0.0236	94.17	0.000	[2.1778, 2.2704]
Male	5.5597	0.1544	35.99	0.000	[5.2569, 5.8625]
Married	4.0145	0.2148	18.68	0.000	[3.5934, 4.4356]
Public sector	1.8446	0.2339	7.89	0.000	[1.3862, 2.3030]
exper	0.1507	0.0081	18.42	0.000	[0.1346, 0.1667]
Urban	3.6681	0.1594	23.01	0.000	[3.3556, 3.9806]
region					

wage	Coefficient	Std. Err.	t	P>|t|	[95% Conf. Interval]
2	4.3575	0.2366	18.41	0.000	[3.8936, 4.8214]
3	1.3068	0.2408	5.43	0.000	[0.8347, 1.7790]
4	0.1666	0.3532	0.47	0.637	[-0.5257, 0.8590]
5	12.1718	0.2668	45.61	0.000	[11.6487, 12.6949]
6	4.1031	0.2646	15.50	0.000	[3.5843, 4.6218]
_cons	-1.6585	0.3922	-4.23	0.000	[-2.4273, -0.8897]
Obs: 178,340	F-stat: 2139.46	R-sq: 0.1171	Root MSE: 31.394		

Table 3: Baseline OLS Regression (Model 1)

MLR.1 & MLR.2 (Linearity & Random Sampling): Satisfied by model design and the use of nationally representative Labour Force Survey (LFS) data.

MLR.3 (No perfect multicollinearity): To ensure that none of the independent variables are perfectly correlated, a Variance Inflation Factor (VIF) test is performed:

Variable	VIF	1/VIF
Education (edu)	2.21	0.4525
Male	1.05	0.9543
Married	1.35	0.7385
Public sector	1.51	0.6611
exper	1.80	0.5563
Urban	1.14	0.8738
Region 2 (Red River)	1.84	0.5435
Region 3 (Central Coast)	1.76	0.5671

Variable	VIF	1/VIF
Region 4 (Central Highlands)	1.26	0.7907
Region 5 (Southeast)	1.65	0.6058
Region 6 (Mekong Delta)	1.65	0.6068
Mean VIF	1.57	

Table 4: Variance Inflation Factor (VIF)

The mean VIF is 1.57, well below the critical threshold of 10. No individual variable exceeds 3. This confirms that multicollinearity is not a serious problem.

MLR.4 (Zero conditional mean): The Ramsey RESET test checks for omitted variable bias and functional form misspecification. - H0: Model has no omitted variables - F(3, 178325) = 1029.68 - Prob > F = 0.0000 The p-value is significant at 1%, meaning we reject the null hypothesis. This indicates potential omitted variable bias (e.g., unobserved innate ability), which is standard in large cross-sectional wage data.

MLR.5 (Homoskedasticity): The Breusch-Pagan test determines if the error variance is constant. - H0: Constant variance - chi2(1) = 70232.58 - Prob > chi2 = 0.0000 We reject the null hypothesis, indicating heteroskedasticity. To correct this violation of MLR.5, heteroskedasticity-robust standard errors are applied in all subsequent regression models.

MLR.6 (Normality of error terms): MLR.6 requires that the error term is normally distributed, which is needed for exact finite-sample t and F inference. However, with a large sample of 178,340 observations, this assumption is not strictly required — by the Central Limit Theorem, OLS estimators are asymptotically normal regardless of the error distribution, ensuring valid inference.

wage	Coefficient	Robust Std. Err.	t	P>|t|	[95% Conf. Interval]
Education (edu)	2.2241	0.0256	86.77	0.000	[2.1738, 2.2743]

wage	Coefficient	Robust Std. Err.	t	P>|t|	[95% Conf. Interval]
Male	5.5597	0.1534	36.22	0.000	[5.2589, 5.8605]
Married	4.0145	0.1802	22.27	0.000	[3.6612, 4.3677]
Public sector	1.8446	0.2864	6.44	0.000	[1.2831, 2.4061]
exper	0.1507	0.0080	18.64	0.000	[0.1348, 0.1665]
Urban	3.6681	0.1471	24.92	0.000	[3.3796, 3.9565]
Region 2 (Red River)	4.3575	0.2267	19.22	0.000	[3.9131, 4.8020]
Region 3 (Central Coast)	1.3068	0.2298	5.69	0.000	[0.8564, 1.7573]
Region 4 (Central Highlands)	0.1666	0.2647	0.63	0.529	[-0.3522, 0.6855]
Region 5 (Southeast)	12.1718	0.2322	52.40	0.000	[11.7166, 12.6271]
Region 6 (Mekong Delta)	4.1031	0.2279	18.00	0.000	[3.6562, 4.5499]
_cons	-1.6585	0.4314	-3.84	0.000	[-2.5041, -0.8130]

Table 5: Baseline Regression with Robust Standard Errors (Model 1 - Robust)

4. Results

Model 1 uses hourly wage (wage) as the dependent variable, while Models 2, 3, and 4 use the natural logarithm of wage (log_wage) to facilitate proportional interpretations. All models are estimated with heteroskedasticity-robust standard errors.

VARIABLES	(1) Model 1 - Linear (wage)	(2) Model 2 - Gender×Public (log_wage)	(3) Model 3 - Edu×Urban (log_wage)	(4) Model 4 - Quadratic (log_wage)
edu	2.226***	0.049***	0.040***	0.050***

VARIABLES	(1) Model 1 - Linear (wage)	(2) Model 2 - Gender×Public (log_wage)	(3) Model 3 - Edu×Urban (log_wage)	(4) Model 4 - Quadratic (log_wage)
married	4.068***	0.127***	0.124***	0.052***
exper	0.149***	0.001***	0.000*	0.023***
urban	3.583***	0.084***	-0.086***	0.078***
region 2	4.358***	0.103***	0.104***	0.106***
region 3	1.185***	-0.012***	-0.009**	-0.010***
region 4	0.213	-0.026***	-0.029***	-0.025***
region 5	11.841***	0.264***	0.264***	0.257***
region 6	4.013***	0.057***	0.055***	0.059***
gender	5.512***	0.181***	0.143***	0.133***
public	1.974***	0.104***	0.007*	-0.005
gender#public	-	-0.197***	-	-
urban#c.edu	-	-	0.015***	-
exper²	-	-	-	-0.000***
Constant	-1.719***	2.647***	2.765***	2.524***
Observations	178,340	178,340	178,340	178,340
R-squared	0.117	0.217	0.215	0.229

*** p<0.01, ** p<0.05, * p<0.1

Table 6: Regression Results

4.1 Interaction term between gender and public sector (Model 2)

To examine the combined effect of gender and sector of employment, an interaction term between gender and public is included in Model 2. This allows us to compare four distinct groups relative to the base group (female workers in the private sector):

- Male in Public sector: 0.181 + 0.104 – 0.197 = 0.088 (approximately +9.2% higher wages)
- Female in Public sector: +0.104 (approximately +10.9% higher wages)
- Male in Private sector: +0.181 (approximately +19.9% higher wages)
- Female in Private sector: Base group (0%)

The results show that the gender wage gap is substantially smaller in the public sector. While male workers enjoy a large premium in the private sector, this advantage is significantly reduced in the public sector. Female workers, on the other hand, benefit more from public sector employment.

4.2 Returns to education: Urban vs Rural workers (Model 3)

Model 3 tests whether the returns to education differ between urban and rural areas by including an interaction term urban#c.edu. The coefficient on edu is 0.040, and the interaction term urban#c.edu is 0.015 (statistically significant at 1% level).

Rural workers: 1 year of additional education increases wages by 4.0%. Urban workers: 1 year of additional education increases wages by 5.5% (0.040 + 0.015).

The F-test for the interaction term rejects the null hypothesis (Prob > F = 0.0000), confirming that education yields significantly higher returns in urban areas. This highlights the importance of location in realizing the benefits of human capital.

4.3 Non-linear relationship between experience and log(wage) (Model 4)

To capture the expected concave relationship between work experience and wages, Model 4 includes a quadratic term ($exper^2$). Both the linear term (exper = 0.023) and the quadratic term ($exper^2$ = -0.0004) are highly significant.

The turning point is calculated as: Peak at 25.48 years of experience ($-\beta_1/(2\times\beta_2)$).

This confirms an inverted U-shaped relationship: wages increase with experience in the early and mid-career stages but begin to decline after approximately 25–26 years, possibly due to skill obsolescence or physical limitations.

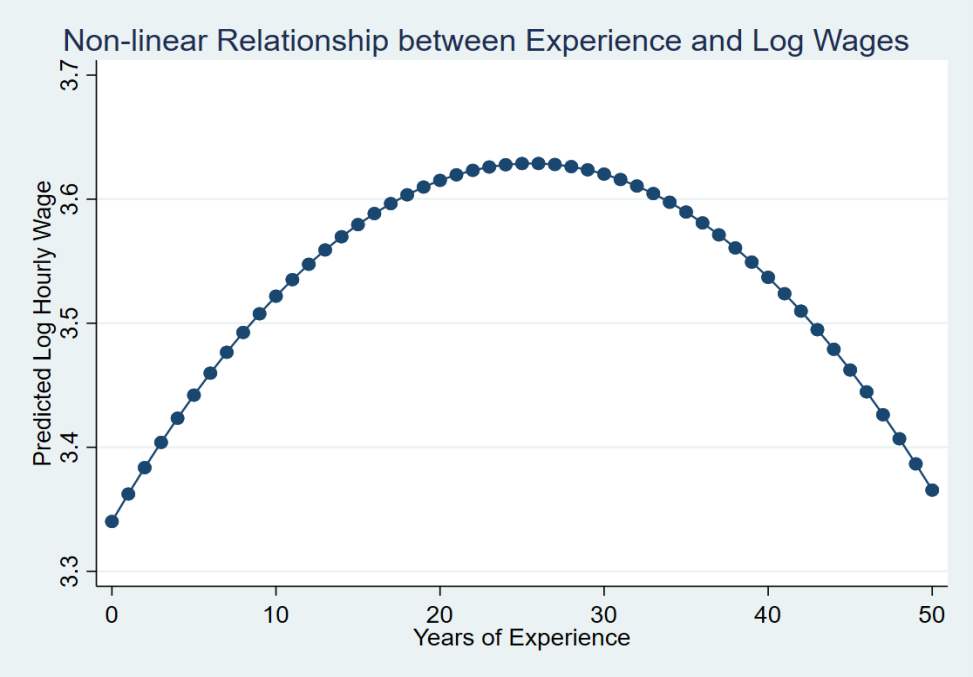


Figure 1: Predicted relationship between experience and ln(wage)

This figure illustrates the relationship between years of work experience and the predicted log of wages. The graph shows that wages increase rapidly during the early stage of the career life cycle. Beyond the turning point, the marginal effect of experience becomes negative, causing the predicted log wage to gradually decline.

4.4 Regional Wage Differentials

Controlling for human capital, demographic, and other job characteristics, the regression results reveal significant wage disparities across regions. The proportional wage differentials relative to the base region (Region 1 - Northern Midlands & Mountains) are calculated using the exact formula for log-linear models: $(\exp(\beta) - 1) \times 100$.

Region	Coefficient	Proportional Differential	% Change	Interpretation
Region 2 (Red River Delta)	0.103	exp(0.103)-1	+10.85%	Higher wages
Region 3 (Central Coast)	-0.012	exp(-0.012)-1	-1.19%	Slightly lower
Region 4 (Central Highlands)	-0.026	exp(-0.026)-1	-2.57%	Lower wages
Region 5 (Southeast)	0.264	exp(0.264)-1	+30.25%	Highest premium
Region 6 (Mekong Delta)	0.057	exp(0.057)-1	+5.87%	Moderately higher

Table 7: Proportional Wage Differentials by Region

The results highlight substantial spatial inequality in Vietnam's labor market. The Southeast region offers the largest wage premium (+30.25%), reflecting its economic dynamism and concentration of high-productivity industries. In contrast, the Central Highlands and Central Coast show wage disadvantages compared to the Northern Midlands & Mountains. These regional disparities persist even after controlling for education, experience, gender, and sector of employment.

5. Conclusion and Policy

5.1 Conclusion

This study confirms that human capital, gender, sector, and geography significantly determine wages in Vietnam. We find that education has a positive effect on wages, with higher returns in urban areas. Gender wage differentials exist, though the gap is substantially smaller in the public sector. Urban-rural and regional wage disparities remain large. Finally, work experience follows an inverted U-shaped relationship with earnings, peaking at roughly 25.5 years.

5.2 Policy implications

Education and Training. The findings suggest that expanding access to education and vocational training in rural areas is essential for reducing wage inequality. Since urban workers tend to receive higher returns to education, improving the quality and accessibility of rural education can help narrow the urban-rural skill premium gap. Investment in human capital development would not only enhance workers' earning potential but also contribute to higher labor productivity and long-term economic growth.

Gender Equality. Policies aimed at promoting gender equality in the labor market should be strengthened, particularly in the private sector where gender wage disparities are more pronounced. Enhancing pay transparency, enforcing equal-pay regulations, and encouraging women's participation in high-skilled and high-paying occupations can help reduce gender-based wage inequality and improve labor market efficiency.

Regional Development. Significant regional wage differences indicate the need for more balanced economic development across Vietnam. Greater public investment and infrastructure development in less-developed regions, such as the Central Highlands, could create better employment opportunities and reduce excessive concentration of economic

activity in the Southeast region. Such measures would help narrow regional income gaps while alleviating migration pressures toward major urban centers.

6. References

Becker, G. S. (1964). *Human Capital*. University of Chicago Press. *https://press.uchicago.edu/ucp/books/book/chicago/H/bo3684031.html*

Becker, G. S. (1985). Human capital, effort, and the sexual division of labor. *Journal of Labor Economics*, *3*(1), S33-S58. *https://doi.org/10.1086/298075*

Blau, F. D., & Kahn, L. M. (2017). The gender wage gap. *Journal of Economic Literature*, *55*(3), 789-865. *https://doi.org/10.1257/jel.20160995*

Gregory, R. G., & Borland, J. (1999). Recent developments in public sector labor markets. In *Handbook of labor economics* (Vol. 3, pp. 3573-3630). Elsevier. *https://doi.org/10.1016/S1573-4463(99)30043-4*

Mincer, J. (1974). *Schooling, Experience, and Earnings*. NBER. *https://www.nber.org/books-and-chapters/schooling-experience-and-earnings*

Vu, T. T. A. (2011). *Decentralization of the economy in Vietnam from an institutional perspective*. *https://papi.org.vn/wp-content/uploads/2011/12/Phan-cap-kinh-te-tai-Viet-Nam-nhin-tu-goc-do-the-che-TS-Vu-Thanh-Tu-Anh.pdf*

World Bank. (2022). *Vietnam poverty and equity assessment 2022: Pathways for inclusive growth and shared prosperity*. *https://openknowledge.worldbank.org/entities/publication/7196058a-c2b2-57ee-b675-f6080ecb0ba5*